AWS EBS Snapshot Lab

Objective
Learn to create an EBS volume, attach it, store data, create snapshots, restore a volume, and verify recovery.

Tasks
1. Create a t2.micro Ubuntu EC2 instance.
2. Create a 5 GB gp3 EBS volume in the same AZ.
3. Attach it as /dev/xvdf.
4. Verify using lsblk.
5. Format: sudo mkfs.ext4 /dev/xvdf
6. Mount at /data and verify with df -h.
7. Create sample files under /data.
8. Create an EBS snapshot named snapshot-lab-backup.
9. Delete the files to simulate data loss.
10. Create a new volume from the snapshot.
11. Detach the old volume and attach the restored one.
12. Mount it again and verify the files are restored.

Questions
What is an EBS snapshot? Are snapshots incremental? Can you create a larger volume from a smaller snapshot? Does deleting a volume delete its snapshots? Explain disaster recovery using snapshots.